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Part 3: Manual Gradient Descent Calculation

Given information:

$$X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$$

$$m = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$$b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

The model is: $\hat{y} = X_m + b$ Learning rate: $\alpha = 0.001$ Setup and update 1

Step 1: Initial prediction

We start with the model

$$\hat{y} = X_m + b$$

First calculation X_m :

$$X_m = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$$

$$X \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

First row:

$$(1 \times -1) + (3 \times 2) = -1 + 6 = 5$$

Second row:

$$(4 \times -1) + (10 \times 2) = -4 + 20 = 16$$

$$X_m = \begin{bmatrix} 5 \\ 16 \end{bmatrix}$$

Now add b:

$$\hat{y} = \begin{bmatrix} 5 \\ 16 \end{bmatrix} +$$

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}$$

Step 2: Error

$$\text{error} = \hat{y} - y$$

$$\text{error} = \begin{bmatrix} 6 \\ 17 \end{bmatrix} -$$

$$\begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$\text{error} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

Step 3: Mean Square Error

$$J = (1/n) \sum (\text{error}^2)$$

$$J = (1/2) (1^2 + 11^2)$$

$$J = (1/2) (1 + 121)$$

$$J = 61$$



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Step 4: Gradients

Gradient formula for m:

$$\partial J / \partial m = (2/n) X^T (\hat{y} - y)$$

Since $n = 2$, then $2/2 = 1$, so:

$$\partial J / \partial m = X^T (\text{error})$$

$$X^T = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix}$$

$$\partial J / \partial m = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \times$$

$$\begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

First value:

$$(1 \times 1) + (4 \times 11) = 45$$

Second value:

$$(3 \times 1) + (10 \times 11) = 113$$

Therefore:

$$\partial J / \partial m = \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

Gradient for b:

$$\partial J / \partial b = \text{error}$$

$$\partial J / \partial b = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

Update parameters

Step 5: Update parameters

Update rules:

$$m_{\text{new}} = m_{\text{old}} - \alpha (\partial J / \partial m)$$

$$b_{\text{new}} = b_{\text{old}} - \alpha (\partial J / \partial b)$$

Update m:

$$m_{\text{new}} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} -$$

$$0.001 \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

$$m_{\text{new}} = \begin{bmatrix} -1 - 0.045 \\ 2 - 0.113 \end{bmatrix}$$

$$m_{\text{new}} = \begin{bmatrix} -1.045 \\ 1.887 \end{bmatrix}$$

$$m_{\text{new}} = \begin{bmatrix} -1.045 \\ 1.887 \end{bmatrix}$$

Update b:

$$b_{\text{new}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} -$$

$$0.001 \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$b_{\text{new}} = \begin{bmatrix} 1 - 0.001 \\ 1 - 0.011 \end{bmatrix}$$

$$b_{\text{new}} = \begin{bmatrix} 0.999 \\ 0.989 \end{bmatrix}$$

$$b_{\text{new}} = \begin{bmatrix} 0.999 \\ 0.989 \end{bmatrix}$$

$$b_{\text{new}} = \begin{bmatrix} 0.999 \\ 0.989 \end{bmatrix}$$

After update 1:

$$m = \begin{bmatrix} -1.045 \\ 1.887 \end{bmatrix}$$

$$b = \begin{bmatrix} 0.999 \\ 0.989 \end{bmatrix}$$

$$b = \begin{bmatrix} 0.999 \\ 0.989 \end{bmatrix}$$

$$b = \begin{bmatrix} 0.999 \\ 0.989 \end{bmatrix}$$

Update 2
Starting values from update 1
 $m = \begin{bmatrix} -1.045 \\ 1.887 \end{bmatrix}$

$$b = \begin{bmatrix} 0.999 \\ 0.989 \end{bmatrix}$$

Step 1: Prediction

$$\hat{y} = xm + b$$

First prediction:

$$\begin{aligned} & (1 \times -1.045) + (3 \times 1.887) + 0.999 \\ &= -1.045 + 5.661 + 0.999 \\ &= 5.615 \end{aligned}$$

Second prediction:

$$\begin{aligned} & (4 \times -1.045) + (10 \times 1.887) + 0.989 \\ &= -4.180 + 18.870 + 0.989 \\ &= 15.679 \end{aligned}$$

$$\hat{y} = \begin{bmatrix} 5.615 \\ 15.679 \end{bmatrix}$$

Step 2: Error

$$\text{error} = \hat{y} - y$$

$$\text{error} = \begin{bmatrix} 5.615 \\ 15.679 \end{bmatrix}$$

$$= \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$\text{error} = \begin{bmatrix} 0.615 \\ 9.679 \end{bmatrix}$$

Step 3: Mean Squared Error

$$J = (1/2)(0.615^2 + 9.679^2)$$

$$J = (1/2)(0.378225 + 93.683041)$$

$$J = 47.031$$

Step 4: Gradient

$$\partial J / \partial m = x^T (\text{error})$$

$$\partial J / \partial m = \begin{bmatrix} 1, 4 \\ 3, 10 \end{bmatrix}$$

X

$$\begin{bmatrix} 0.615 \\ 9.679 \end{bmatrix}$$

first value:

$$\begin{aligned} & (1 \times 0.615) + (4 \times 9.679) \\ &= 0.615 + 38.716 \\ &= 39.331 \end{aligned}$$

second value:

$$\begin{aligned} & (3 \times 0.615) + (10 \times 9.679) \\ &= 1.845 + 96.790 \\ &= 98.635 \end{aligned}$$

$$dJ/dm = \begin{bmatrix} 39.331 \\ 98.635 \end{bmatrix}$$

Gradient for b:

$$dJ/db = \begin{bmatrix} 0.615 \\ 9.679 \end{bmatrix}$$

Step 5: update parameters

update m:

$$m_{\text{new}} = \begin{bmatrix} -1.045 \\ 1.887 \end{bmatrix}$$

$$-0.001 \begin{bmatrix} 39.331 \\ 98.635 \end{bmatrix}$$

$$m_{\text{new}} = \begin{bmatrix} -1.045 - 0.039331 \\ 1.887 - 0.098635 \end{bmatrix}$$

$$m_{\text{new}} = \begin{bmatrix} -1.084331 \\ 1.788365 \end{bmatrix}$$

update b:

$$b_{\text{new}} = \begin{bmatrix} 0.999 \\ 0.989 \end{bmatrix}$$

$$-0.001 \begin{bmatrix} 0.615 \\ 9.679 \end{bmatrix}$$

$$b_{\text{new}} = \begin{bmatrix} 0.999 - 0.000615 \\ 0.989 - 0.009679 \end{bmatrix}$$

$$b_{\text{new}} = \begin{bmatrix} 0.998385 \\ 0.979321 \end{bmatrix}$$

after update 2:

$$m = \begin{bmatrix} -1.084331 \\ 1.788365 \end{bmatrix}$$

$$b = \begin{bmatrix} 0.998385 \\ 0.979321 \end{bmatrix}$$

Starting values from Update 2

$$m = [-1.084331, 1.788365]$$

$$b = [0.998385, 0.979321]$$

$$\text{Prediction: } \hat{y} = X_m + b$$

First prediction:

$$\begin{aligned} & (1 \times -1.084331) + (3 \times 1.788365) + 0.998385 \\ &= -1.084331 + 5.365095 + 0.998385 \\ &= 5.279149 \end{aligned}$$

Second prediction:

$$\begin{aligned} & (4 \times -1.084331) + (10 \times 1.788365) + 0.979321 \\ &= -4.337324 + 17.883650 + 0.979321 \\ &= 14.525647 \end{aligned}$$

$$\text{Error: } \hat{y} - y$$

$$\text{error} = [[5.279149], [14.525647]] - [[5], [6]]$$

$$\text{Error} = [[0.279149], [8.525647]]$$

Mean Squared Error

$$J = (1/2)(0.279149^2 + 8.525647^2)$$

$$J \approx 36.382$$

Gradients

$$\frac{\partial J}{\partial m} = X^T (\text{error})$$

$$\frac{\partial J}{\partial m} = [[1, 4], [3, 10]] \times [[0.279149], [8.525647]]$$

$$\begin{aligned} \text{First value: } & (1 \times 0.279149) + (4 \times 8.525647) \\ &= 34.381737 \end{aligned}$$

$$\begin{aligned} \text{Second value: } & (3 \times 0.279149) + (10 \times 8.525647) \\ &= 86.093917 \end{aligned}$$

$$\frac{\partial J}{\partial m} = [[34.381737], [86.093917]]$$

$$\text{Gradient for } b: \frac{\partial J}{\partial b} = [[0.279149], [8.525647]]$$

Update Parameters:

Update m:

$$m_{\text{new}} = [-1.084331, 1.788365] - 0.001 [34.38137, 26.01317]$$

$$m_{\text{new}} = [-1.084331 - 0.03438137, \\ 1.788365 - 0.02601317]$$

$$m_{\text{new}} = [-1.118713, 1.702271]$$

Update b:

$$b_{\text{new}} = [0.998385, 0.979321]$$

$$- 0.001 [0.279149, 8.525647]$$

$$b_{\text{new}} = [0.998106, 0.970795]$$

After Update 3:

$$m = [-1.118713, 1.702271]$$

$$b = [0.998106, 0.970795]$$

Update 4 and final trend

Starting Variables from update 3

$$m = \begin{bmatrix} -1.118713, \\ 1.702271 \end{bmatrix}$$

$$b = \begin{bmatrix} 0.998106, \\ 0.970795 \end{bmatrix}$$

Step 1 : Prediction

$$\hat{y} = X_m \cdot b$$

First Prediction:

$$\begin{aligned} & (1 \times -1.118713) + (3 \times 1.702271) + 0.998106 \\ &= -1.118713 + 5.106813 + 0.998106 \\ &= 4.986206 \end{aligned}$$

Second Prediction:

$$\begin{aligned} & (4 \times -1.118713) + (10 \times 1.702271) + 0.970795 \\ &= -4.474852 + 17.022710 + 0.970795 \\ &= 13.518653 \end{aligned}$$

$$\hat{y} = \begin{bmatrix} 4.986206, \\ 13.518653 \end{bmatrix}$$

Step 2 : Error

$$\text{error} = \hat{y} - y$$

$$\text{error} = \begin{bmatrix} 4.986206, \\ 13.518653 \end{bmatrix}$$

$$= \begin{bmatrix} 5, \\ 6 \end{bmatrix}$$

$$\text{error} = \begin{bmatrix} -0.013794, \\ 7.518653 \end{bmatrix}$$

Step 3 : Mean Squared Error

$$J = \left(\frac{1}{2} \right) ((-0.013794)^2 + 7.518653^2)$$

$$J = 28.265$$

Step 4 : Gradients

$$\partial J / \partial m = X^T (\text{error})$$

$$\begin{aligned} \partial J / \partial m &= \begin{bmatrix} 1, 4, \\ 3, 10 \end{bmatrix} \\ &\times \begin{bmatrix} -0.013794, \\ 7.518653 \end{bmatrix} \end{aligned}$$

First Value:

$$\begin{aligned}(1 \times -0.013794) + (4 \times 7.618653) \\ = -0.013794 + 30.474612 \\ = 30.460818\end{aligned}$$

Second Value

$$\begin{aligned}(3 \times -0.013794) + (10 \times 7.618653) \\ = -0.041382 + 75.186530 \\ = 75.145148 \\ \partial J / \partial m = \begin{bmatrix} 30.460818, \\ 75.145148 \end{bmatrix}\end{aligned}$$

Gradient for b:

$$\partial J / \partial m = \begin{bmatrix} -0.013794, \\ 7.618653 \end{bmatrix}$$

Step 5: Update Parametres

Update m:

$$\begin{aligned}m_{\text{new}} &= \begin{bmatrix} 1.218713, \\ 1.702271 \end{bmatrix} \\ &- \\ &0.001 \begin{bmatrix} 30.460818, \\ 75.145148 \end{bmatrix}\end{aligned}$$

$$m_{\text{new}} = \begin{bmatrix} -1.118713 - 0.030460818, \\ 1.702271 - 0.075145148 \end{bmatrix}$$

$$m_{\text{new}} = \begin{bmatrix} -1.148774, \\ 1.627126 \end{bmatrix}$$

Update b:

$$\begin{aligned}b_{\text{new}} &= \begin{bmatrix} 0.998106, \\ 0.970795 \end{bmatrix} \\ &- \\ &0.001 \begin{bmatrix} -0.013794, \\ 7.618653 \end{bmatrix}\end{aligned}$$

$$b_{\text{new}} = \begin{bmatrix} 0.998106 + 0.000013794, \\ 0.970795 - 0.007618653 \end{bmatrix}$$

$$b_{\text{new}} = \begin{bmatrix} 0.998120, \\ 0.963276 \end{bmatrix}$$

After Update 4:

$$m = \begin{bmatrix} -1.148774, \\ 1.627126 \end{bmatrix}$$

$$b = \begin{bmatrix} 0.998120 \\ 0.963276 \end{bmatrix}$$

Final Trend explanation

The mean Squared error decreases as the updates continue. It starts at 61, then decreases to approximately 47.031, 36.382 and 28.265. This shows that gradient descent is working correctly because the values of m and b are moving in a direction that reduces the prediction error. Each update adjusts the parameters in the opposite direction of the gradient, helping the predictions move closer to the actual y values.

SUMMARY TABLE

Stage	m_1	m_2	b_1	b_2	MSE
Initial	-1.000000	2.000000	1.000000	1.000000	61.000
After Update 1	-1.045000	1.887000	0.999000	0.989000	47.031
After Update 2	-1.084331	1.788365	0.998385	0.979321	36.382
After Update 3	-1.118713	1.702271	0.998016	0.970995	28.265
After Update 4	-1.148774	1.627126	0.998120	0.963276	22.077

NB, The MSE shown for "After update 4" is calculated using the final m and b after the fourth update. The detailed section above shows the calculation that produced the first update.